Chemical Engineering Journal 486 (2024) 150196

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Ionic-confining-assisted multiple-mode tunable light emitting of carbon nanodots

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ARTICLE INFO

Keywords:
Carbon nanodots
Solid-state luminescence
Phosphorescence
Chemiluminescence
Information security

ABSTRACT

Multiple-mode light emitting in one material system is tremendous desirable for various applications. Herein, we have developed an ionic crystal confining strategy to prepare the carbon nanodots (CDs) with quadruple-mode light emitting and demonstrated their promising application in information anti-counterfeiting via optical logic gates. In detail, the quadruple-mode light emitting of CDs (QM-CDs) are prepared with the citric acid and urea with NaOH matrix via microwave-assisted polymerization. Mechanically, the high-density ionic bonds stemmed from the NaOH ionic-crystal can effectively inactivate nonradiative defect centers, stabilize triplet excitons, increase absorption cross-section and boost resonance energy transfer, evoking the outstanding solid-state fluorescence, upconversion photoluminescence (UCPL), phosphorescence, and chemiluminescence simultaneously. Meanwhile, the isolated NaOH matrix can promote the confining aggregation of CDs, leading to the redshifted emission wavelength and endowing the light emitting with tunable wavelength. With the excitability of logic functions between the light emitting and controlled excitation, the programmable multiple-inputs logic gates are established with the QM-CDs, enabling the application in high-throughput logical operations and information anti-counterfeiting. This research provides a new insight into the synthesize of multiple-mode luminescent CDs, and thus promote the applications of nanomaterials in intelligent encryption and anti-counterfeiting.

1. Introduction

Optical information encryption can realize advanced anti-counterfeiting by taking advantages of the multiple light emitting properties, including the excitation and emission of wavelength, intensity, lifetime, etc [1–8]. With the conventional approach to optical input and output signals, the decrypted information derived from light emission wavelength, intensity and lifetime are greatly restricted by the requirement of characterization equipment, spectral overlap or background interference [3,7–9]. As an information dimension, the excitation and emission mode also can be coded as the basic input and output signals because they are independent of the accurate spectral information, thus avoiding the requirement of complex characterization devices and interference of spectral overlap and background noise. Recently, there have been several reports about the strategy of encryption and decryption through the light emitting under different excitation mode, enabling the smart information anti-counterfeiting via simple naked-eyes [10,11]. However, the most research on optical information encryption are focused on the combination of fluorescence and phosphorescence, or fluorescence and chemiluminescence (CL) [10–15]. There are rare researches about the anti-counterfeiting technology with the combination of multiple-mode light emitting. With the actual situation of accurate excitation mode and photophysical process, the simultaneous multiple-mode light emitting is difficult in one materials system. Therefore, it is highly desired to design and upgrade the multiple-mode light emitting of materials, and thus promote the optical anti-counterfeiting technology.

Carbon nanodots (CDs), as one new class of carbon-based nanomaterials, have attracted extensive attention due to their high photoluminescence quantum yield (PL QY), tunable emission wavelength, high photostability and excellent biocompatibility [10,16–21]. To date, various CDs have been prepared with conjugate or non-conjugate

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https://doi.org/10.1016/j.cej.2024.150196
Received 31 December 2023; Received in revised form 16 February 2024; Accepted 4 March 2024
Available online 5 March 2024
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molecules [14,22–29]. And multiple modal light emitting, including fluorescence, phosphorescence, upconversion photoluminescence (UCPL), and CL, has been realized with the CDs as fluorophore through external light injection, multiphoton absorption, triplet-exciton transitions, and energy transfer [30–37]. Nevertheless, there are still several obstacles to realize the effective light emitting simultaneously in one CD systems

(1) nonradiative decay and quenching centers. Due to the inappropriate push and pull electron-group incorporation and heteroatom substitutions or doping in CDs, defect centres can easily form and lead to the irreducible nonradiative recombination. [38–41] (2) Small UCPL absorption cross-section. Unlike rare earths with rich transition electron energy levels, the CDs, which are constructed in carbon skeleton with sp^2 conjugate domain, possess relatively small UCPL absorption cross-sections [26,42–44]. (3) Triplet-exciton quenching. Since the triplet excitons are known to dissipate by nonradiative transitions, delayed fluorescence, triplet–triplet annihilation, and luminescence bursts, the phosphorescence from CDs always suffers serious triplet-excitons quenching. [23,31,33,41] (3) Useless energy transfer. Since resonance energy transfer can easily occur when the absorption of acceptor is overlap with the emission of donor, there are few reports about the effective chemiluminescence resonance energy transfer (CRET) between CDs and CL donors. [14,45,46] Therefore, it is significant to develop the CDs with outstanding multiple-mode light emitting simultaneously and ensure their promising application.

In this work, we have developed an ionic crystal confining strategy to prepare the CDs with quadruple-mode light emitting and demonstrated their promising application in information anti-counterfeiting via optical logic gates. In detail, the quadruple-mode emitting of CDs (QM-CDs) are prepared with the microwave-assisted polymerization of citric acid and urea in NaOH aqueous solution, and exhibit tunable solid-state fluorescence (SF), UCPL, phosphorescence, and CL simultaneously in one material system. Mechanically, the high-density ionic bonds stemmed from the NaOH ionic-crystal can effectively inactivate nonradiative defect centers, stabilize triplet excitons, increase absorption cross-section and boost resonance energy transfer, evoking the outstanding SF, UCPL, phosphorescence, and CL. Meanwhile, the isolated NaOH matrix can promote the confining aggregation of CDs, leading to the redshifted emission wavelength and endowing the light emitting with tunable wavelength. With the excitability of logic functions between the light emitting and controlled excitation of the QM-CDs, we have established a programmable multiple-inputs logic gates and achieved the high-throughput logical operations and information. This research paves a new sight to the light emitting of CDs, and thus promote the applications of nanomaterials in intelligent encryption and anti-counterfeiting.

2. Experiment section

2.1. Materials

Citric acid ($C_6H_8O_7$, 99 %), urea (CON_2H_4, 99.9 %), sodium hydroxide (NaOH, 97 %), luminol ($C_8H_7N_3O_2$, 97 %), cobalt chloride hexahydrate ($CoCl_2{\cdot}6H_2O$, 97 %) and tris-(hydroxymethyl) aminomethane ($C_3H_{11}NO_3$, 99 %) were purchased by Macklin Chemistry Co. Ltd (Shanghai, P. R. China). Hydrogen peroxide (H_2O_2, 30 %) was purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, P. R. China). The chemicals reagents were analytical grade and used as received without further purification.

2.2. Characterization

The morphology was characterized by a field emission transmission electron microscope (HRTEM, JEOL JSM-IT100). The UV–vis absorption was measured on a Hitachi U-3900 UV–VIS-NIR spectrophotometer. The fluorescence, phosphorescence and chemiluminescence spectra were measured by a spectrofluorometer (Hitachi F-7000). The phosphorescence and chemiluminescence lifetimes were measured by a spectrofluorometer (Hitachi F-7000). The X-ray photoelectron spectroscopy (XPS) was measured on a Kratos AXIS HIS 165 spectrometers with a monochromatized Al KR X-ray source (1486.7 eV). The Fourier transform infrared spectra (FT-IR) were recorded on a Bio-Rad Excalibur spectrometer (Bruker vector 22). The XRD patterns was measured with the X-ray diffractometer (Panalytical X' Pert Pro) using Cu k_α as the irradiation source. The Raman spectra were carried on a Raman spectrometer (Renishaw in Via). The absolute PL QYs were measured with FLS1000 spectrometer, and fluorescence lifetimes were characterized using FLS1000 time-corrected single photon counter system. A regenerative amplified femtosecond Ti

Sapphire laser system (Coherent: 800 nm, 70 fs, 6 mJ per pulse), and 1 kHz repetition rate was used to collect the upconversion photoluminescence.

2.3. Synthesis of the CDs and QM-CDs

For the CDs, 1.8 g citric acid and 2.5 g urea were dissolved in 30 mL deionized water and stirred for 30 min. The obtained clear solution was heated in a microwave oven (750 W) for 3 min. After the samples were cooled down to room temperature, 20 mL ethanol was mixed with the products and the obtained solution was centrifuged (8000 rpm, 10 min) to remove the precipitates. Finally, the solid CDs powder was obtained by freeze-drying. For the QM-CDs, 1.8 g citric, 2.5 g urea and 1.1 g NaOH were dissolved in 20 mL deionized water and sonicated for 5 min, and then the QM-CDs are prepared as same as the CDs. For comparation, different CDs are prepared with 1.8 g citric, 2.5 g urea and 0.3 g / 0.5 g / 0.7 g/ 0.9 g NaOH.

2.4. CL emission measurement of the CDs and QM-CDs

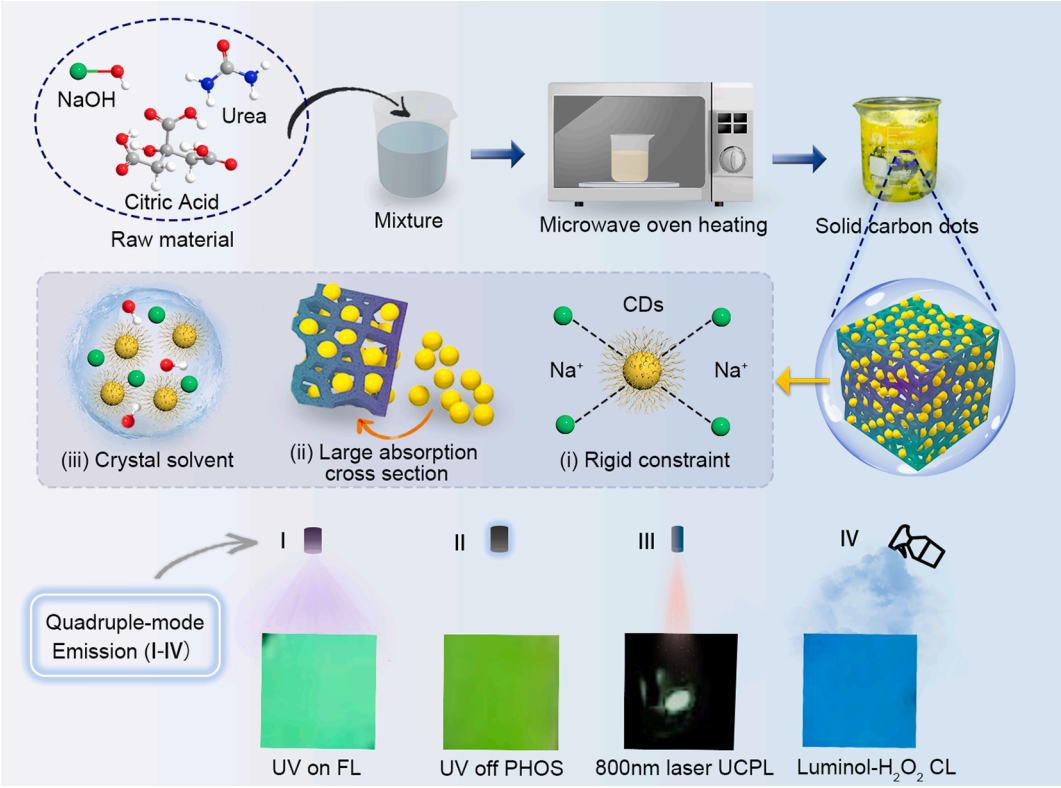
The CL spectra were measured with a Hitachi F-7000 FL spectrophotometer with the light source off. 6 mM luminol solution was prepared by dissolving luminol, 20 mM $CoCl2{\cdot\cdot}6H_2O$ and 25 mM tris-(hydroxymethyl) aminomethane in 10 mM NaOH aqueous solution. The working solutions of H_2O_2 were freshly prepared from 30 % H_2O_2 (1 M). The luminol/H_2O_2 solution was injected into the QM-CDs powder and the CL kinetic curves were collected after the solution mixing with an F-7000 FL spectrophotometer.

2.5. DFT computational details

The geometries of the ground state structures of HPPT and its HPPT-Na^+ complex was fully optimized without any structural constraints by density functional theory (DFT) in Gaussian 09 package. The frequency calculations were performed by the same level of theory to confirm that the optimized ground-state geometry was a stable structure. Time-dependent DFT (TD-DFT) calculations were carried out for the vertical excitation energies of both singlet and triplet excited states. The PBE0-D density functional (including the D3 version of Grimme's dispersion) and def2-TZVP basis set were employed in the calculation. The spin–orbit coupling matrix elements were calculated by TD-DFT calculation as implemented in ORCA program.

3. Results and discussion

Since the excitons are known to dissipate by nonradiative transitions, triplet–triplet annihilation, and luminescence bursts, two prerequisites are essential to improve the light emitting capability of fluorophores [31,39,40,47,48]. One is to boost the excitons by accelerating the exciton recombination, and another is to minimize the non-radiative transition. With these prerequisites, we propose one facile ionic crystal confining strategy to achieve the quadruple-mode emitting in a CDs system by constructing a rigid molecular environment [31]. As illustration in Scheme 1, the source of citric acid, urea and NaOH are



Scheme 1. Schematic illustration for the preparation of quadruple-mode emitting CDs.

employed to prepare the CDs and the in-situ ionic-crystal matrix can embrace around the CD units to provide the rigid molecular environment. With the ionic-crystal matrix, the formation of high-density ionic bonds in the QM-CDs can effectively passivate the nonradiative defect centers, stabilize the triplet excitons, increase the cross-sectional area of the UCPL absorption, and boost the CRET between the CD units and luminol-H_2O_2 reaction, endowing the CDs with outstanding SF, UCPL, phosphorescence and CL. Compared with the individual pure source and microwave-treated source, the as-prepared QM-CDs present far higher PL and phosphorescence emission intensity (Figure S1 and S2), verifying the importance of microwave-assisted polymerization for the light emitting of QM-CDs.

To understand the ionic crystal confining effect, the unconfined CDs are prepared with the microwave-assisted polymerization of citric acid and urea. As shown in Fig. 1a and 1b, the transmission electron microscopy (TEM) image of the CDs and QM-CDs present clear dot shape with a size distribution ranging from 1.5 to 5.0 nm, respectively. Meanwhile, the ionic-crystal matrix can be observed in the TEM image of QM-CDs and the high-resolution TEM (HRTEM) image of the CDs and QM-CDs reveals the same lattice spacing of 0.21 nm attributed to the (100) crystallographic plane of graphitic carbon. Meanwhile, the X-ray diffraction (XRD) patterns of the CDs and QM-CDs are also compared. As illustrated in Fig. 1c, the CDs only present a main peak at around 27.1° attributed to the (110) crystallographic plane of sp^3-hybrid carbon nitride while the (002) crystallographic plane of graphitic carbon and diffraction peaks attributed to (200), (400), (311) and (020) crystallographic plane of monoclinic NaOH crystals emerge and in the QM-CDs, further verifying the different frameworks of CDs and QM-CDs, and the formation of NaOH matrix [31,39,51]. The Raman spectra of the CDs and QM-CDs exhibit the same shifts at 1347.0 cm^{-1} attributed to D peak and at 1579.0 cm^{-1} attributed to G peak (Fig. 1d), verifying the similar sp^2/sp^3 carbon cores in the framework of CDs and QM-CDs [49]. The approximate I_G/I_D ratios of 1.0 for the CDs and 1.1 for the QM-CDs imply that the QM-CDs possess higher graphitization degree in carbon core. Similarly, the Fourier transform infrared (FTIR) spectra of the CDs and QM-CDs exhibit the same peaks at 3447.8 cm^{-1} attributed to the $\nu_{(O—H/N—H)}$, 1673.5 cm^{-1} attributed to the $\nu_{(C=O/C=N)}$ and 1615.3 cm^{-1} attributed to the $\nu_{(C=C)}$ (Fig. 1e), verifying the similar surface functional groups for the CDs and QM-CDs [50,51]. In the survey of X-ray photoelectron spectroscopy (XPS), the CDs and QM-CDs exhibit the same elements of carbon (~246 eV), nitrogen (~399 eV) and oxygen (~532 eV) while the QM-CDs present another element of Na (~1072 eV) (Fig. 1f and 1 g). The similar element ratios of C1s, N1s and O1s confirm the similar element component of the fluorophores in the CDs and QM-CDs. For the CDs and QM-CDs, the C1s spectrum can be deconvoluted into C—C/C=C (284.6 eV), C—O/C—N (286.4 eV) and C=O (288.6 eV) (Fig. 1h and 1j), the N1s spectra can be pyridine N (397.7 eV) and pyrrole N (399.1 eV) (Fig. 1i and 1 k) and the O1s can be carbonyl C=O (531.6 eV) and C—O (532.2 eV) (Figure S3), verifying their similar carbon cores and surface functional groups, corresponding to the analysis of TEM image, XRD pattern, FTIR and Raman spectra. With the survey of C1s spectra, the percentage of sp^2 C are 47.1 % for the CDs and 50.2 % for the QM-CDs, implying the high higher graphitization degree of QM-CDs. And the QM-CDs present higher content of graphitic N, further confirming the higher graphitization degree of the QM-CDs. Summarizing all these results, it can be evidently revealed that the QM-CDs are composed of the ionic-confined CDs with a rigid monoclinic NaOH environment and the introduced NaOH matrix can promote the graphitization degree of CDs.

Whereas, the SF, UCPL, phosphorescence, and CL emission of the CDs and QM-CDs are compared. As illustrated in Fig. 2a, the QM-CDs powder can emit bright SF under 365 nm UV lamp while there is nearly no fluorescence from the pure CDs powder. The PL excitation-emission matrix reveals that the QM-CDs can emit excitation-independent fluorescence under the excitation wavelength ranging from 300 nm to 470 nm with two centers at (374 nm, 512 nm) and (420 nm, 513 nm) (Fig. 2b). With the direct measurement, the solid-state PL QY of the QM-CDs can be calculated as high as 38.38 % (Figure S4). Similarly, the QM-

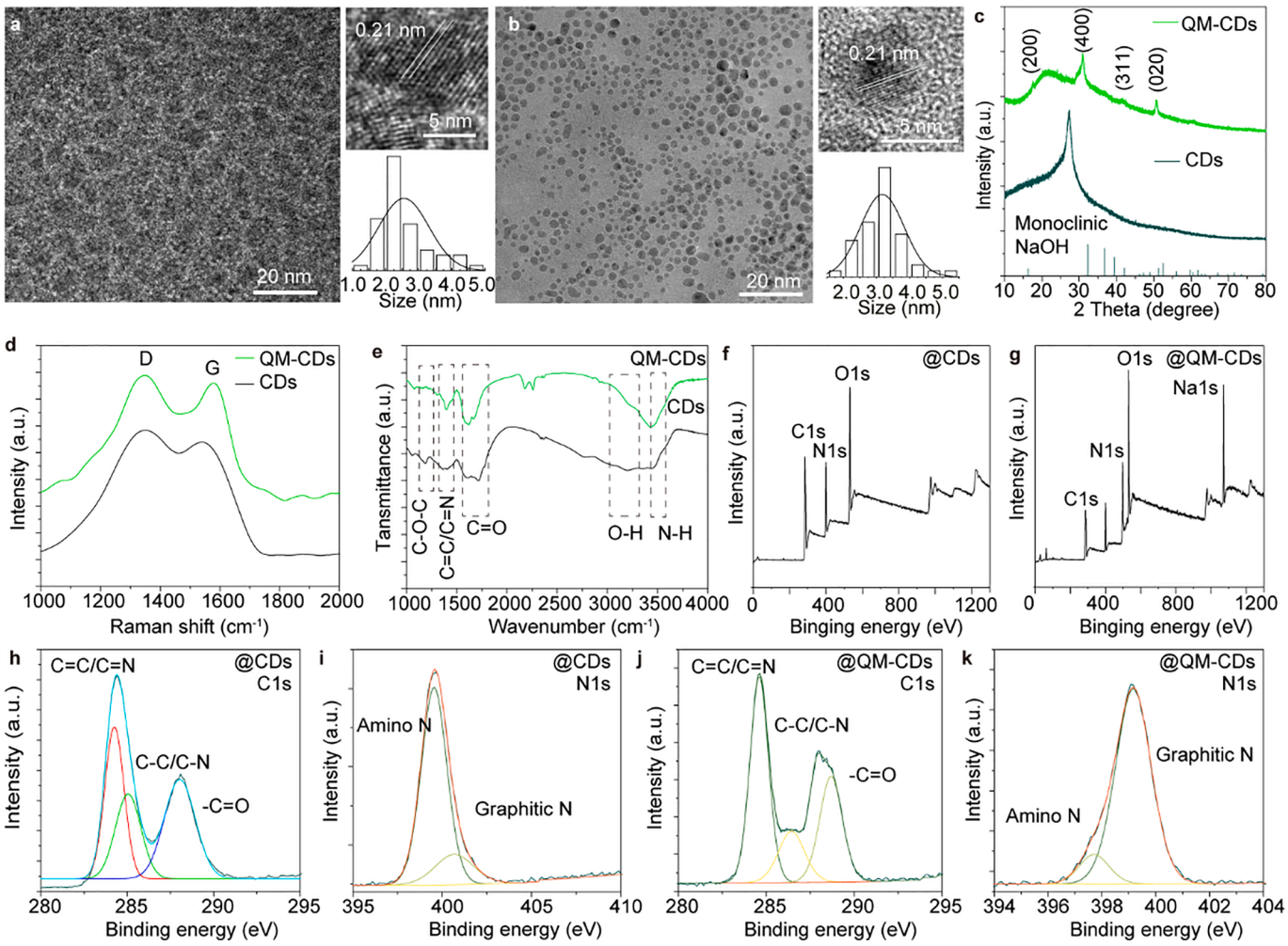


Fig. 1. a)-b) tem images of the cds (a) and qm-cds (b) (inset: high-resolution tem images of the cds and qm-cds, and their size distribution). c) xrd patterns of the cds and qm-cds. d) raman spectra of the qm-cds and cds. e) ftir spectra of the cds and qm-cds. f)-g) full xps survey of the cds (f) and qm-cds (g). h)-i) high-resolution c1s (h) and n1s (i) xps survey of the cds. j)-k) high-resolution c1s (j) and n1s (k) xps survey of the qm-cds.

CDs powder also can emit excellent UCPL emission. As shown in Fig. 2c, the QM-CDs powder can emit 530 nm photon under 800 nm femtosecond laser, which are much higher than the CDs powder. With the reference of Rhodamine 6G (Figure S5 and Table S1), the absorption cross-section (σ) of the QM-CDs can be calculated as 5.3 GM, which is higher than that of 1.7 GM for the CDs (Fig. 2d) [26]. Meanwhile, the QM-CDs powder present long-persistent phosphorescence after UV lamp off (Fig. 2e). As illustrated in Fig. 2f, the QM-CDs can emit excitation-independent phosphorescence and the excitation-emission matrix exhibits two canters at (280 nm, 504 nm) and (345 nm, 510 nm). The phosphorescence spectra of the QM-CDs powder present nearly same performance as their PL emission under the maximal excitation (Fig. 2g), implying the ionic crystal confining-assisted triplet exciton emission as previous report. With the decay curve, the average afterglow lifetime of the QM-CDs can be calculated as 47.27 ms (Fig. 2h). With the PL emission and phosphorescence spectra (Figure S6 and Table S2), the absolute phosphorescence QY can be calculated as 6.83 % and the corresponding intersystem crossing (ISC) rate is $4.93 \times 10^6\ s^{-1}$. In addition, the CDs and QM-CDs can achieve efficient energy transfer in luminol-H_2O_2 reaction. As shown in Fig. 2i, bright cyan CL emission can be observed after adding the CDs or QM-CDs into the luminol-H_2O_2 solution. Compared with the CL peak at 425 nm for the initial luminol-H_2O_2 reaction and 448 nm for the CDs, the QM-CDs present the CL peak at 478 nm, verifying that the ionic-crystal confining can effectively boost the CRET. Meanwhile, the QM-CDs illustrate stable CL spectra and the QM-CDs-based CL lifetime can be calculated as 60.8 s (Fig. 2j and 2 k). With the CL decay and the reference of lucigenin-H_2O_2 CL reaction (Figure S7), the CL QY of the QM-CDs in luminol- H_2O_2 solution is 6.01×10^{-3} einsteins mol^{-1}, which is quite lower than that of 3.65×10^{-2} einsteins mol^{-2} for the initial luminol- H_2O_2 reaction and 1.63×10^{-2} einsteins mol^{-1} for the CDs in luminol- H_2O_2 solution (Fig. 2j) [24,25]. With the overlap between the absorption spectra of QM-CDs aqueous solution and the CL spectra of luminol-H_2O_2 reaction, the CRET efficiency can be calculated as high as 75.0 % (Figure S8).

To understand the ionic crystal confining effect, the photophysical properties of the QM-CDs are further evaluated with the density functional theory (DFT) calculation. As illustrated in Fig. 3a, the fluorescent aromatic ring can generate in the QM-CDs from citric acid and urea molecules through dehydration and carbonization. With the aromatic ring model, the absorption and PL emission of the QM-CDs can be calculated (Fig. 3b). As shown in Fig. 3c, UV–vis absorption and PL emission of the QM-CDs aqueous solution present the similar performance as the calculated results. And the UV–vis absorption of QM-CDs aqueous solution present abundant overlap with the CL emission of luminol-H_2O_2 reaction, enabling the efficient resonance energy transfer. To gain insight into the light emission mechanism, the energy level distributions of the CDs and QM-CDs are further estimated. As shown in Fig. 3d and 3e, the fluorescent aromatic ring models with and without Na + are established and the excited state energies, spin–orbit couplings (ξ) and natural transition orbitals are calculated. The results reveal that

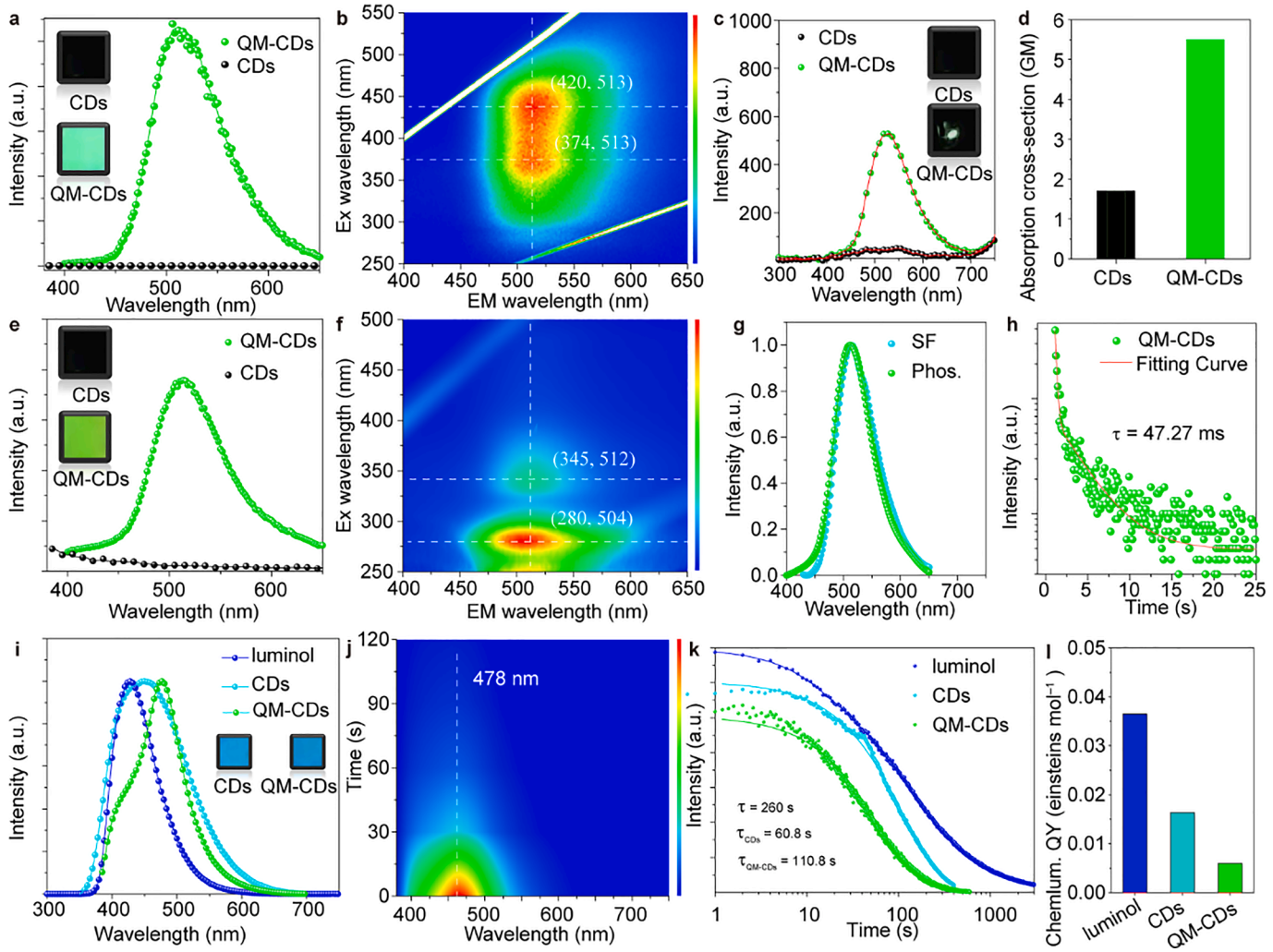


Fig. 2. a) pl emission spectra of the cds and qm-cds (inset: photographs of the cds and qm-cds under 365 nm UV lamp). b) PL excitation-emission matrix of the QM-CDs. c) UCPL spectra of the CDs and QM-CDs (Inset: photographs of the CDs and QM-CDs under 800 nm laser). d) UCPL absorption cross-section of the CDs and QM-CDs. e) Phosphorescence spectra of the CDs and QM-CDs (Inset: photographs of the CDs and QM-CDs after UV lamp off). f) Phosphorescence excitation-emission matrix of the QM-CDs. g) PL and phosphorescence emission spectra of the QM-CDs under their maximal excitation. h) Time-resolved decay of the QM-CDs after light off. i) CL spectra of the initial luminol-H_2O_2 and after adding the CDs and QM-CDs (Inset: photographs of the CDs and QM-CDs in luminol-H_2O_2). j) Time-resolved CL spectra of the QM-CDs in luminol-H_2O_2. k) CL intensity decay of the luminol-H_2O_2, CDs and QM-CDs. l) CL QY of the luminol-H_2O_2, CDs and QM-CDs in luminol-H_2O_2.

the spin orbit coupling (SOC) ξ (S_1, T_n) of the QM-CDs (16.91 cm^{-1}) is larger than that of the unconfined CDs (8.901 cm^{-1}), indicating that the high-density ionic bonds can effectively passivate the nonradiative defect centers and stabilize the triplet excitons as previous report. Hence, with the model of confining isolated CDs, the CL emission of the QM-CDs can be further understood. As shown in Fig. 3f, the theoretical calculation reveals that the highest occupied molecular orbital (HOMO) level of luminol (−5.64 eV) is lower than that of the QM-CDs (−6.72 eV) and CDs (−6.47 eV), but higher than the lowest unoccupied molecular orbital (LUMO) level of the QM-CDs (−2.89 eV) and CDs (−2.31 eV), indicating the feasible energy transfer between the luminol and QM-CDs or CDs. On the basis, the quadruple-mode light emitting mechanism of the QM-CDs can be deduced. As illustration in Fig. 3g, the initial CD particles will undergo serious aggregation-caused quenching (ACQ) due to nonradiative transition such as energy transfer, surface electron recombination and π-π interactions in the aggregation state, endowing the low SF to the CDs. In the presence of NaOH matrix, the additional NaOH can improve the crystallinity degree of carbon core and reduce the formation of defects in CD. Meanwhile, the NaOH matrix can break down the CDs into individual cluster by amide bond hydrolysis, allowing the formation of ionic bonds and avoiding the resonance energy transfer [31]. Whereupon, the QM-CD powder present excellent SF under light illumination. Similarly, with the high-density ionic bonding interactions between CD and ionic crystals, the QM-CDs can provide larger cross-section for UCPL absorption. Thereupon, the QM-CDs illustrate far higher UCPL intensity than the CDs. Meanwhile, the high-density ionic bonds in the QM-CDs allow the NaOH matrix to form a rigid molecule network, which can effectively suppress the nonradiative transition caused by molecular vibrations and promote the ISC between excited singlet and triplet, resulting in the long-persistent phosphorescence emission. For the luminol-based CL emission, the double negative ion (Dianion) can be produced from the luminol-H_2O_2 reaction and further oxidized by the oxygen decomposed from H_2O_2 to form an organic peroxide, and the unstable peroxide immediately decomposes to produce N_2 gas and give rise to the excited 3-aminophthalic acid. Thus, the excited-state production can transit and resulting in the 425 nm CL emission in the initial luminol-H_2O_2 reaction (Figure S9) [14,45]. Thus, when the CDs or QM-CDs are added into the luminol-H_2O_2 solution, the CRET can occur between luminol molecules and fluorophore in the QM-CDs and result in the excitation of fluorophore, endowing the glare CL

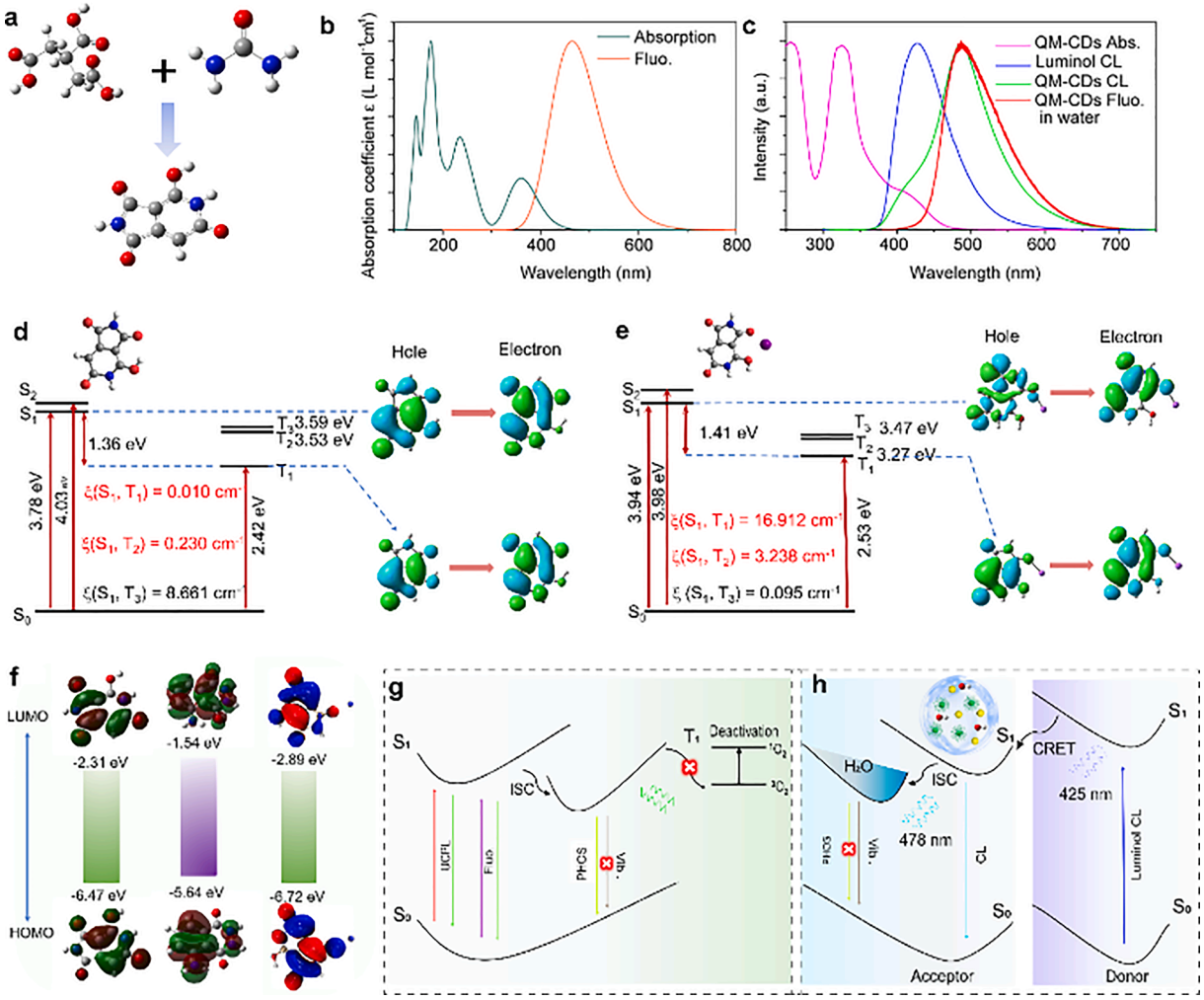


Fig. 3. a) schematic illustration of the fluorescent aromatic ring in the qm-cds. b) calculated pl emission spectra and uv–vis absorption of the fluorescent aromatic ring. c) PL emission spectra and UV–vis absorption of the QM-CDs aqueous solution, and the CL emission spectra of the luminol-H_2O_2 and QM-CDs. d)-e) Calculated excitation energies, spin–orbit couplings (ξ) and natural transition orbitals for the CDs (d) and QM-CDs (e). (f) Calculated HOMO and LUMO distributions of the luminol molecules, QM-CDs, and CD fluorophore. g)-h) Schematic illustration of the light emission mechanism for the PL, UCPL, phosphorescence (g) of the QM-CDs under light excitation, and the CRET between the QM-CDs and luminol-H_2O_2 reaction (h).

emission.

In addition, the products obtained from the direct heat treatment, solvothermal reaction and physical mixture of pure CDs and NaOH are compared with the QM-CDs. With the measurements, it can be revealed that the direct heat treatment, solvothermal reaction and physical mixture can decrease the ACQ effect of CDs and achieve quite bright solid-state fluorescence (Figure 10). However, these obtained powder presents far lower phosphorescence emission intensity than the QM-CDs and the CL emission from these CDs without microwave treatment are nearly same as the pure luminol-H_2O_2 reaction, clearly demonstrating that the microwave reaction is necessary for the formation of QM-CDs. Meanwhile, the NaOH content are removed from the QM-CDs via the dialysis in aqueous solution, and the PL, phosphorescence and CL emission of the obtained CDs powder present similar PL emission spectra as the initial QM-CDs while the obtained CDs powder present lower phosphorescence intensity and a slight redshift emission peak as the pure luminol-H_2O_2 reaction (Figure S11), clearly demonstrating that the NaOH ionic-crystal can effectively inactivate nonradiative defect centers, stabilize triplet excitons, increase absorption cross-section and boost resonance energy transfer, further evoking the outstanding SF, UCPL, phosphorescence, and CL. Moreover, the LiCl-CDs, NaCl-CDs and KCl-CDs are prepared with the source of citric acid and urea in LiCl, NaCl, KCl via microwave-assisted polymerization as the QM-CDs (Figure S12), respectively. The obtained LiCl-CDs, NaCl-CDs and KCl-CDs present quite bright fluorescence but far weaker phosphorescence than the QM-CDs, implying that the weak ionic bonds stemmed from the salts of LiCl, NaCl and KCl can only slightly evoke the phosphorescence of CDs. And the LiCl-CDs, NaCl-CDs and KCl-CDs present same CL emission as the initial luminol-H_2O_2 reaction, implying that the weak ionic bonds stemmed from the salts of LiCl, NaCl and KCl cannot promote the CRET between the CDs and luminol-H_2O_2 reaction. Similarly, the MPD-CDs prepared with m-phenylenediamine (MPD), urea and sodium hydroxide, and the CA-CDs prepared with citric acid and sodium hydroxide present similar bright PL emission and phosphorescence emission and different CL emission as the initial luminol-H_2O_2 solution (Figures S14 and S15), implying that only the high-density ionic bonds stemmed from the NaOH can effectively evoke the outstanding solid-state fluorescence, phosphorescence, and CL simultaneously. Moreover, the RhB@NaOH from microwave-assisted polymerization of RhB and NaOH, and the ZnO@NaOH from microwave-assisted polymerization of ZnO QDs and NaOH also present bright PL emission and phosphorescence emission, also demonstrating the universality of ionic crystal confining strategy for the light emission of molecules and quantum dots (Figure S17 and S18). Compared with the pure CDs, the ionic crystal confining can further boost the energy transfer between CDs and luminol, enabling the longer CL emission wavelength. With the

experiment characterizations and theorical calculation, it can be revealed that the high-density ionic bonds stemmed from the ionic-crystal in the QM-CDs can effectively inactivate the nonradiative defect centers, stabilize the triplet excitons, increase the absorption cross-section of UCPL and boost the energy transfer in CL reaction, evoking the efficient multiple-mode light emitting.

On the other hand, the photophysical properties of the QM-CDs can be further regulated by tuning the reactant content of NaOH. As illustrated in Fig. 4a, the CDs prepared with different NaOH content illustrate different SF, phosphorescence, and CL emission. With the increased NaOH content from 0.3 g to 1.1 g, the SF peak of CDs gradually decreases from 550 nm to 500 nm (Fig. 4b), the phosphorescence peak gradually decreases from 571 nm to 520 nm (Fig. 4c), and the CL peak gradually decreases from 538 nm to 478 nm (Fig. 4d). And the relative yield of CDs increases from 46.6 % to 65.2 % when the NaOH content increase from 0.3 g to 1.1 g (Figure S19 and Table S3) and the prepared QM-CDs powder present excellent repeatability (Figure S20 and Table S4)., implying the potential industrialization production of CDs. The spread wavelength confirm that the multicolor light emitting can be easily achieved (Fig. 4e). Meanwhile, the QM-CDs prepared with different NaOH content also present different phosphorescence lifetime. As illustrated in Fig. 4f, with the increased NaOH content from 0.3 g to 1.1 g, the phosphorescence lifetime of the QM-CDs gradually increases from 0.1259 ms to 47.27 ms. Similarly, the solid-state PL QY of QM-CDs powder increase from 0.39 % to 36.76 % (Figure S21) and the PL QY of QM-CDs aqueous solution increase from 30.06 % to 68.54 % as the amount of NaOH increase from 0.3 g to 1.1 g (Figure S22 and Table S5), indicating that the NaOH ionic-crystal can effectively inactivate non-radiative defect centers and decrease the aggregation-caused-quenching (ACQ) effect. In these QM-CDs, the photophysical properties of the QM-CDs prepared with 0.3 g NaOH present the similar performance as the QM-CDs prepared with 1.1 g NaOH (Figure S23). To understand the alteration in photophysical properties, the QM-CDs are further dispersed in water for comparation. As shown in Fig. 4g, compared with the PL and phosphorescence emission of the QM-CDs powder, the PL emission of the QM-CDs aqueous solution exhibits a 22 nm blueshift, verifying the possible aggregation-caused redshift as previous report [31]. Meanwhile, the UV–vis absorption of the QM-CDs exhibits two absorption bands around 450 nm and 650 nm while the QM-CDs only present the absorption bands around 450 nm and 320 nm (Fig. 4h), implying the confining-caused aggregation. In addition, the QM-CDs powder and aqueous solution exhibits the PL lifetime of 5.69 ns and 4.32 ns (Fig. 4i), further confirming the interparticle interaction of CD fluorophore in the monoclinic NaOH. Therefore, it can be deduced that the in-situ confining isolated aggregation in the QM-CDs result in the red-shift PL, phosphorescence and CL emission of the similar fluorescent CD

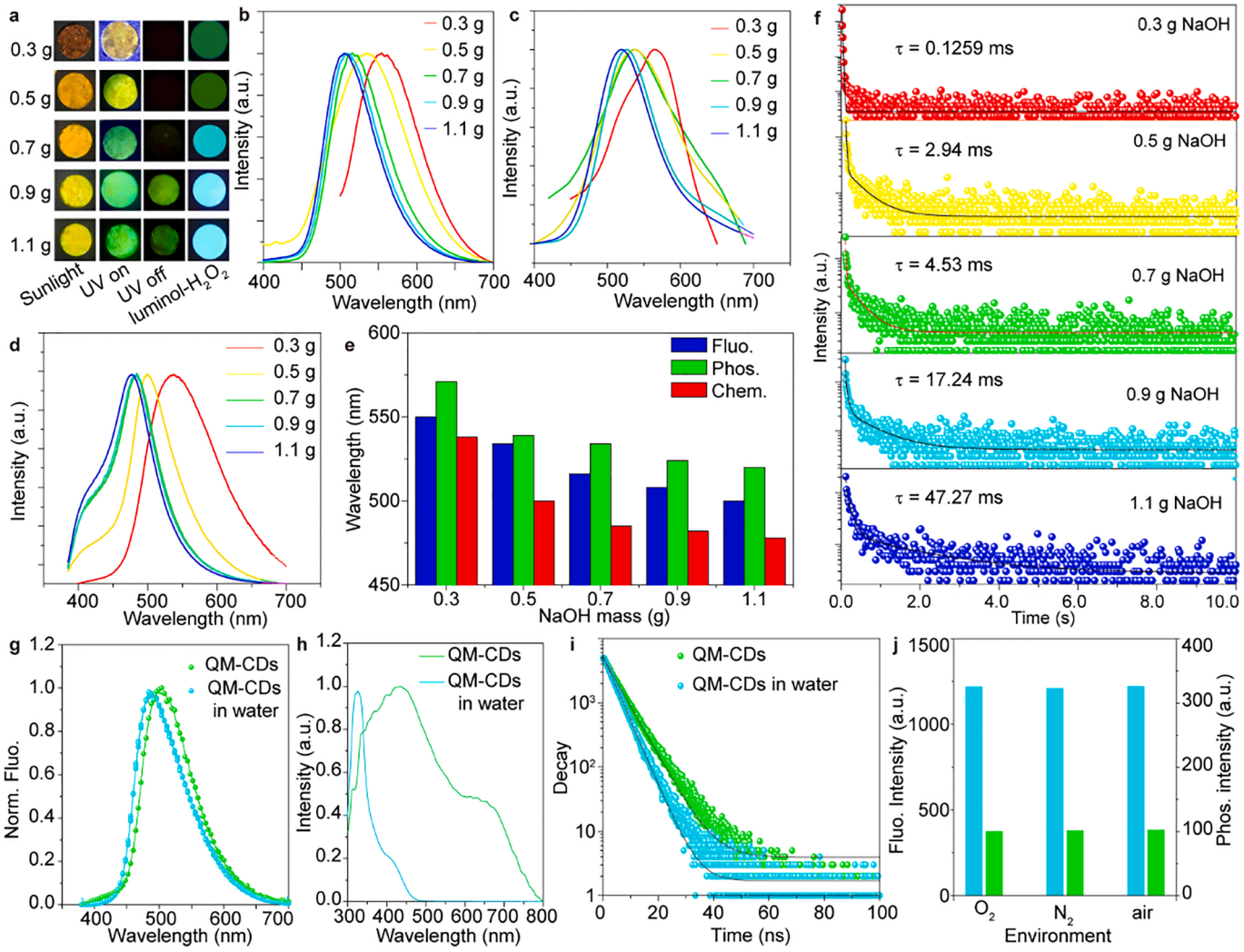
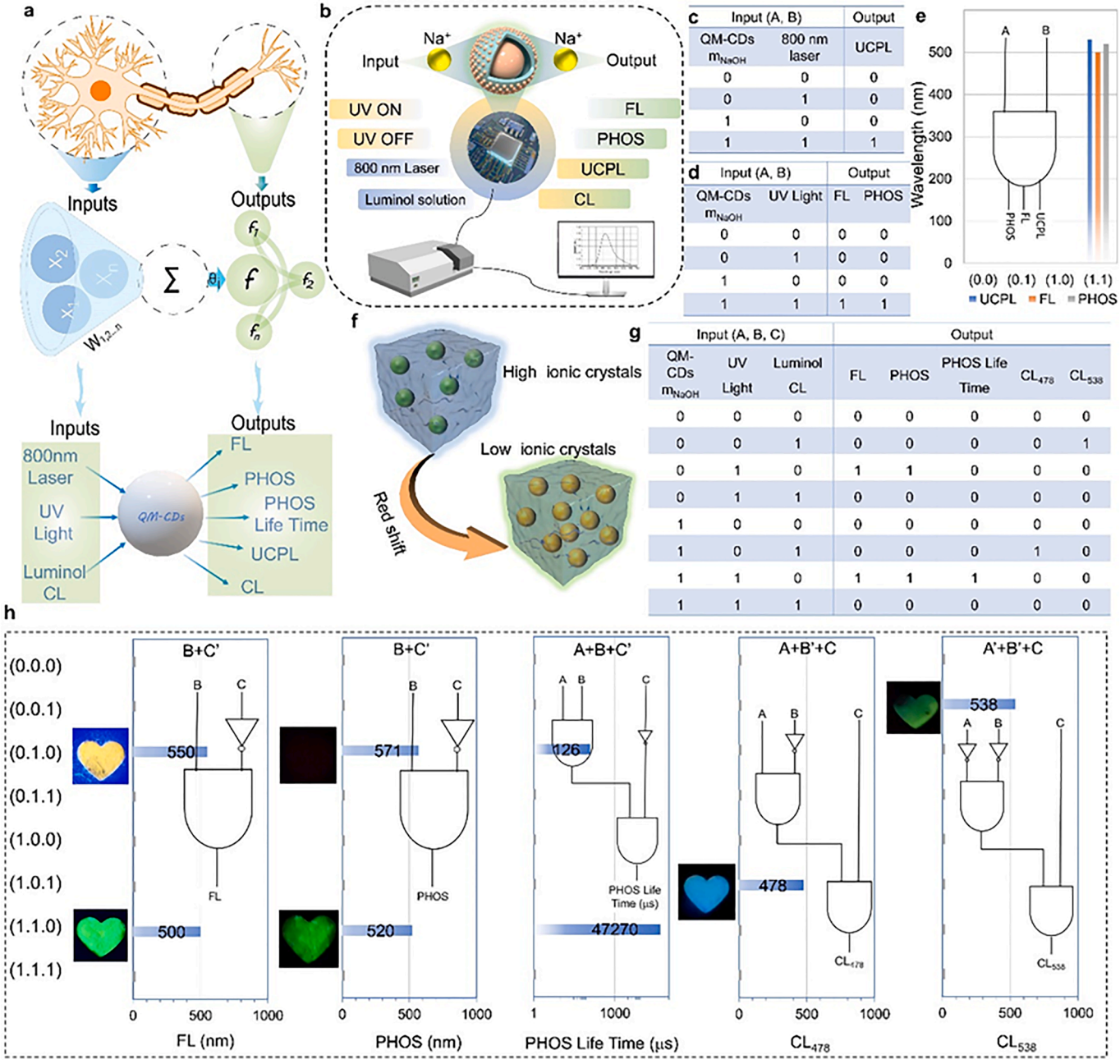


Fig. 4. a) optical pictures of the qm-cds prepared with different naoh content under sunlight, uv lamp, uv lamp off and in luminol-h_2O_2 solution. b)-d) SF (b), phosphorescence (c) and CL (d) emission spectra of these QM-CDs. e) The SF, phosphorescence and CL emission peak of these QM-CDs. f) The time-resolved phosphorescence intensity of these QM-CDs. g)-i) The PL emission spectra (g), UV–vis absorption (h), and PL decay (i) of the QM-CDs and QM-CDs in water. j) SF and phosphorescence intensity of the QM-CDs in N_2, O_2 and air environment.

particles. Moreover, the QM-CDs powder illustrate stable SF and phosphorescence emission. As shown in Fig. 4j, the QM-CDs powder illustrate stable SF and phosphorescence in N_2, O_2 and air atmosphere (Fig. 4j), verifying the isolated CD fluorophore in the rigid molecular environment. Similarly, when the QM-CDs are stored in N_2, O_2 and CO_2 for 1 day, the powder present nearly the same PL intensities as in the air, directly confirming the little influence of the possible adsorption of moisture and carbon dioxide (Figure S24). Meanwhile, the PL emission spectra of the QM-CDs prepared with different NaOH content remain almost unchanged after more than 11 days, clearly demonstrating the stability of QM-CDs in air (Figure S25). Similarly, the QM-CDs present excellent photostability, in which the SF and phosphorescence emission of QM-CDs present almost unchanged after more than 60 min illumination of 365 nm UV lamp (Figure S26). In addition, when the QM-CDs are stored in air with different relative humidity, the QM-CDs present quite stable PL emission in which the QM-CDs only present slight decrease in PL emission intensity for the relative humidity ranging from 10 % to 70 % (Figure S27).

The multicolor light emitting promotes us to employ the QM-CDs as mediate to achieve information encryption. As schematic illustrated in Fig. 5a, the QM-CDs-based logic gate are constructed with the excitation inputs (e. g. UV, 800 nm laser, luminol-H_2O_2 solution) and emission outputs (e. g. light emitting wavelength, phosphorescence lifetime). In the logic gates, the emission outputs are dependent on the excitation



Input (A, B)		Output
QM-CDs m_{NaOH}	800 nm laser	UCPL
0	0	0
0	1	0
1	0	0
1	1	1

Input (A, B)		Output	
QM-CDs m_{NaOH}	UV Light	FL	PHOS
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Input (A, B, C)			Output				
QM-CDs m_{NaOH}	UV Light	Luminol CL	FL	PHOS	PHOS Life Time	CL_{478}	CL_{538}
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	1
0	1	0	1	1	0	0	0
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0
1	1	0	1	1	1	0	0
1	1	1	0	0	0	0	0

Fig. 5. a) schematic illustration of the qm-cds-based logic gate with different excitation inputs (e. g. uv, 800 nm laser, luminol-H_2O_2) and emission outputs (e. g. light emitting, phosphorescence lifetime). b) Schematic illustration of the dual-input and single/dual-output logic gates. c) -d) The truth table of AND gate logic gates based on the NaOH content and excitation mode. e) Histogram of the output data in the QM-CDs-based logic gates. f) Schematic illustration of the regulated QM-CDs for logic gate. g) The truth table of three-input and multi-output logic gates based on three signal response changes in QM-CDs with NaOH content, UV light, and luminol-H_2O_2. h) The schematic plots for the truth table, histogram output data, and constructed logic gate.

input and determined by the photophysical properties of QM-CDs. Thus, the inputs and output can be programmed by excitation modes and emission characters (Fig. 5b). Herein, different excitation modes can be selected to stimulate the logic gates, and the output signals are collected by the spectrometer to distinguish the information data. With the messages efficiently delivered in seconds with the QM-CDs, the unique logical nature can be used to perform logical operations. In practical, the NaOH content in QM-CDs as well as the excitation modes are employed to act as inputs, and thus the emission signals of SF, phosphorescence (PHOS), UCPL and CL can act as outputs, which can be defined as "1″ and "0″ for the presence and absence of NaOH, UV light, 800 nm laser or luminol-H_2O_2 solution. The generated emission signals as output signals in different excitation modes can be recorded. For examples, with the QM-CDs prepared with 1.1 g NaOH and the presence 800 nm laser in the logic gate, the input (A, B) can be defined as input (1, 1) and the single output of "0″ can obtained in the logic systems (Fig. 5c). Similarly, for the QM-CDs without NaOH and the presence of UV light, the input (A, B) can be defined as input (0, 1), and dual-output of (0, 0) can be obtained from the FL and PHOS emission (Fig. 5d). As shown in Fig. 5e, by modulating the stimulation of the logic gates under different excitation modes, the light emission of 530 nm UCPL, 500 nm SF, and 520 nm PHOS can be defined as output of "1" or "0", confirming the logic operations. Moreover, with the input of the NaOH mass, UV light and luminol- H_2O_2 solution as inputs, the outputs of *FL*, *PHOS*, *PHOS* lifetime, *CL* emission can be obtained and defined from the photophysical information (Fig. 5f–h and S8). On the basis, a logic system with three inputs and five outputs can be developed. Moreover, the complex logic operations are performed (B AND NOT C, for FL, PHOS), ((A AND B) AND C, for PHOS lifetime), ((A AND NOT B) AND C, for CL_{478}), and ((NOT A AND NOT B) AND C, for CL_{538}). With the logic systems, the information can be encrypted and decrypted with complex strategy.

Since the QM-CDs-based logic gates possess the logical capability, it is promising to employ them in the advanced information anti-counterfeiting. Usually, the encryption (cryptography), hiding (steganography), and anti-access (password protection) are three common methods in the information protection. When the information is encrypted, it is usually converted into unintelligible cipher text. The steganography and cryptography are advanced forms of cryptography that combine the cryptography with steganography as encryption-steganography. Hence, with the partial fragments of the QM-CDs neuromorphic models and the programmability of QM-CDs, the advanced molecular steganography and cryptography can be construct with the logical computations. As schematic illustration in Fig. 6a, a joint model of encryption and steganography has been established for basic steganography, in which the secret message can be encrypted or decrypted based on the secret key. With the help of a steganographic secret key, the object can also be decrypted. Based on the definition of the logical "0/1″ of the molecular logic systems, eight combinations of three inputs can output eight lines of five-bit binary code and the hidden information can be decrypted through steganalysis. In the process, the encrypted information is firstly encoded into the cover or carrier in some way, and then the eight lines of five-bit binary code can be output on five information channels from the input to the output in the QM-CDs-based neuromorphic model (Fig. 6b and c). Meanwhile, with the necessary decrypted secret key, the five-bit secret key can be decrypt into the two secret messages (88.8E, 38.4 N) and (84.8E, 38.8 N) (Fig. 6d), respectively. On the condition, the decoding of the hidden information is accomplished by steganography. Based on the clues provided by the secret messages, the decrypted secret messages (88.8E, 38.4 N) and (84.8E, 38.8 N) represent the latitude/longitude for further decryption. From these

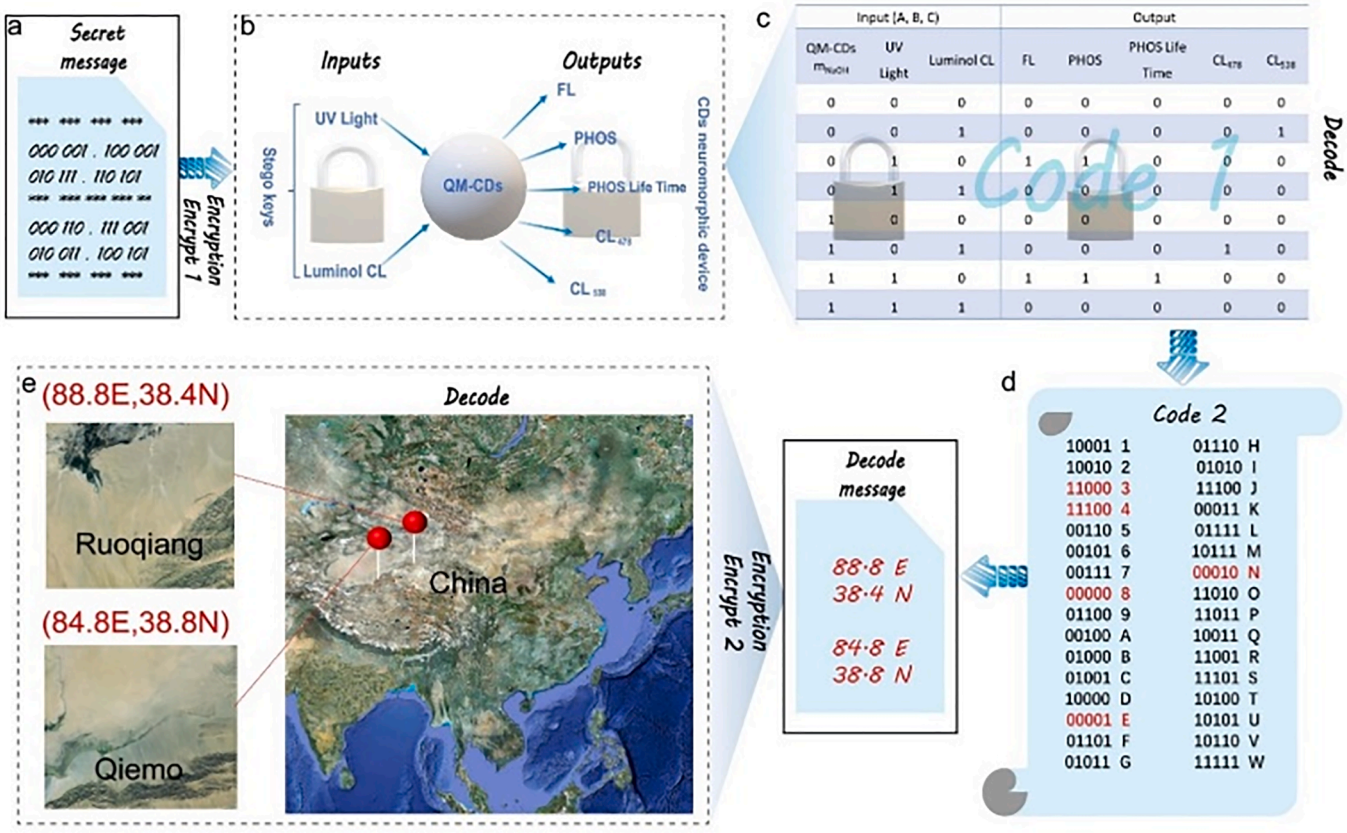


Fig. 6. a) steganography with the information in a cover/carrier. b) the morphological models of neurons in the qm-cds receives and senses 3 inputs (as stego keys) as stego objects to produce multi-response outputs of wavelength, and lifetimes. c) the corresponding input–output binary combinations which can be decoded to obtain 5-bit hidden messages. d) The corresponding 5-bit key from the decoded messages. e) The decrypted coded messages of (88.8E, 38.4 N) and (84.8E, 38.8 N) and their meanings (latitude and longitude signals) displayed by map positioning.

hidden and encrypted messages, a place in Ruoqiang and a place in Qiemo can be identified (Fig. 6e). With the unique molecular hiding and programmability, the QM-CDs-based neuron-mimicking logic gates can enable the molecular cryptography and have great potential for facilitating information encryption and security.

4. Conclusions

In conclusion, the quadruple-mode emissive CDs with tunable SF, UCPL, phosphorescence, and CL emission are prepared with the microwave polymerization citric acid, urea and NaOH. The experimental characterizations and theoretical calculations confirm that the formation of ionic bonds can passivate the non-radiative defect centers, stabilize triplet excitons, boost UCPL absorption cross-section, and enhance the CRET. And the isolated NaOH matrix can promote the confining aggregation of CDs and lead to the redshifted emission wavelength. With the multiple-mode light emitting, the programmable logic gates are established and achieve the high-throughput logical operations and information anti-counterfeiting. This work paves a new sight into the light emitting mechanism of CDs, and thus provide a facile strategy to design and synthesize more promising nanomaterials.

CRediT authorship contribution statement

Jing-Nan Hao: Methodology, Investigation, Formal analysis. **Zhong-Zheng Ding:** Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Cheng-Long Shen:** Writing – original draft, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Guang-Song Zheng:** Project administration, Methodology, Formal analysis, Data curation. **Jia-Lu Liu:** Methodology, Investigation, Formal analysis, Data curation. **Ruo-Bing Li:** Formal analysis, Data curation. **Run-Wei Song:** Methodology, Investigation, Formal analysis, Data curation. **Kai-Kai Liu:** Project administration, Methodology, Investigation. **Jin-Hao Zang:** Formal analysis, Data curation. **Lin Dong:** Validation, Supervision, Methodology, Investigation. **Qing Lou:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization. **Chong-Xin Shan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

We gratefully acknowledge the support of the National Natural Science Foundation of China (12261141661, 12074348, 11974317 and 62204223), the China Postdoctoral Science Foundation (2022TQ0307), and the Natural Science Foundation of Henan Province (212300410078 and 222102310664).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.150196.

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